

SANGUNG PARK

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|  [SCOPUS ID: \[hazsang@outlook.com\]\(mailto:hazsang@outlook.com\) / Sangung Park](#)
Vancouver, BC - V6S 0M2, Canada
11/11/2025 Updated.

RESEARCH INTERESTS

KEYWORDS DIRECTLY RELATED TO THE POSITION ARE UNDERLINED.

- **Smart city**, complex systems, system dynamics, and agent-based models,
- **Disaster science** and large-scale human mobility data analysis, and
- **Transportation Planning**, transportation systems, and accessibility analysis.

EDUCATION

- **Purdue University** Aug 2019 - Aug 2024
Ph. D. in Lyles School of Civil Engineering West Lafayette, US
 - GPA: 3.72/4.00
 - Dissertation title: Critical Transitions of Post-disaster Recovery via Data-driven Interdependent Socio-physical Multi-agent Systems
 - Studied and worked at Urban Mobility Networks and Intelligence Lab with Dr. Satish V. Ukkusuri.
- **University of Seoul** Feb 2016 - Aug 2017
M.S. in Transportation Engineering Seoul, South Korea
 - GPA: 4.38/4.50
 - Thesis title: Traffic Flow Analysis under a Connected and Autonomous Environment through Market Penetration Rates
 - Studied and worked at the Transportation Planning Lab with Dr. Seungjae Lee.
- **University of Seoul** Mar 2010 - Feb 2016
B.S. in Transportation Engineering and Statistics Seoul, South Korea
 - GPA: 3.53/4.00
 - Project title: Transit Scheduling of Subway No. 9 to Reduce Congestion

EXPERIENCE

- **University of British Columbia**  Sep 2025 - Current
Postdoctoral Research Fellow Vancouver, Canada
 - Research at the Multimodal Mobility Systems lab advised by Dr. Amy Kim.
 - Research on the accessibility analysis of the communities in British Columbia.
- **University of Florida**  Aug 2024 - Aug 2025
Postdoctoral Associate Gainesville, USA
 - Research at SERMOS lab advised by Dr. Xilei Zhao.
 - Lead a project as a Principal Investigator, "Developing a Dynamic Social Vulnerability Index for Public Health" (Total \$ 5,000).
 - Support TSIS-CORSIM 3D visualization and marketing strategies at the McTrans Center.
 - Advise students:
 - * Graduate students: Thomas Florian Veit and Shangkun Jiang.
 - * Undergraduate students: Ally Recalde and Matthew Mapa.
- **Seoul Institute**  Jul 2017 - Jun 2019
Researcher Seoul, South Korea
 - Conduct several projects:
 - * A Study on Efficient Operation of a Plan on a Special Transportation and Expansion of low-floor bus
 - * A Study on an Improvement Plan on a Bicycle Environment
 - * A Base Study on the Introduction of New Transportation to Improve Traffic Congestion and Vitalize Tourism
 - * A Special Plan on the Green Transportation Promotion Area in Seoul.
 - * A Base Study on the Restructuring of Gwanghwamun Square in Seoul.

SELECTED RESEARCH PROJECTS

- **A deep learning approach for detecting built environment in transit-oriented developments** Aug 2024 - May 2025
Sponsor: UTC - Center for Equitable Transit Oriented Communities (CETOC) Role: Researcher
- **Developing a dynamic social vulnerability index (DSVI) for public health using location-based data in Hurricane Beryl (\$5,000)** Oct 2024 - Apr 2025
Sponsor: Public Health Extreme Events Research (PHEER) network Role: PI
 ◦ PI: Dr. Sangung Park
- **Analyzing transit-based evacuation demand in hurricanes** Aug 2024 - May 2025
Sponsor: Center for Equitable Transit Oriented Communities (CETOC) Role: Researcher
- **Cybersecurity Testbed for Connected and Autonomous Vehicles (CAVs)** Jan 2024 - May 2024
Sponsor: USDOT National Center for Transportation Cybersecurity and Resiliency (TraCR) Role: RA
- **Adapting Transport Systems for Future Heat Extremes** Aug 2023 - Dec 2024
Sponsor: World Bank - Global Facility for Disaster Reduction and Recovery Role: RA
- **CRISP Type2: Critical Transitions in the Resilience and Recovery of Interdependent Social and Physical Networks** Aug 2019 - Dec 2022
Sponsor: National Science Foundation Role: RA
 ◦ Collaborations with social science (Dr. Seungyoon and Dr. Laura Siebeneck) and electrical and computer engineering (Dr. Shreyas Sundaram).
- **A Study on Efficient Operation of a Plan on a Special Transportation and Expansion of low-floor bus.** Jul 2018 - Jun 2019
Sponsor: Ministry of Land, Infrastructure and Transport, South Korea. Role: Researcher
- **A Study on an Improvement Plan on a Bicycle Environment.** Jan 2018 - Dec 2018
Sponsor: Seoul Metropolitan Government, Seoul, South Korea. Role: Researcher
- **A Special Plan on the Green Transportation Promotion Area in Seoul.** Jan 2018 - Dec 2018
Sponsor: Seoul Metropolitan Government, Seoul, South Korea. Role: Researcher
- **A Base Study on the Restructuring of Gwanghwamun Square in Seoul.** Jul 2017 - Dec 2018
Sponsor: Seoul Metropolitan Government, Seoul, South Korea. Role: Researcher
- **A Study on the Autonomous Driving System Using Traffic Data-oriented Data Mining and V2X Communication Technologies.** Jul 2015 - Jul 2017
Sponsor: National Research Foundation of Korea, South Korea. Role: RA

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, T=THESIS, O=OTHERS

- [O.1] Ukkusuri, S. V., Mittal, S., **Park, S. U.**, Jones, N., Avner, P., Romero, N. (2025). [Heatwaves and Their Effects on Transportation Systems: A Comprehensive Review](#). *World Bank*.
- [J.12] Xue, J., **Park, S.**, Mondal, W. U., Reia, S. M., Yao, T., and Ukkusuri, S. V. (2024). [An agent-based model of post-disaster recovery in multilayer socio-physical networks](#). *Sustainable Cities and Society*, Vol. 115, pp. 105863. DOI: 10.1016/j.scs.2024.105863
- [J.11] **Park, S.**, Yabe, T., and Ukkusuri, S. V. (2024). [Post-disaster recovery policy assessment of urban socio-physical systems](#). *Computers, Environment, and Urban Systems*, Vol. 114, pp. 102184. DOI: 10.1016/j.compenvurbsys.2024.102184
- [J.10] Ukkusuri, S. V., **Park, S.**, Mittal, S., Chapman, L., Manoli, G., Santos, A., Jones, N. K. W., Avner, P., Romero, N., et al. (2024). [We need to prepare our transport systems for heatwaves—here’s how](#). *Nature*, Vol. 632, Issue 8024, pp. 253-256. DOI: 10.1038/d41586-024-02538-8
- [J.9] **Park, S.**, Kim, M., and Kim, J. (2024). [Hourly Long-term Traffic Volume Prediction with Meteorological Information using Graph Convolutional Networks](#). *Applied Sciences*, Vol. 14, Issue 6, pp. 2285. DOI: 10.3390/app14062285.
- [J.8] **Park, S.**, Yao, T., and Ukkusuri, S. V., (2024). [Spatiotemporal Heterogeneity Reveals Urban-Rural Differences in Post-disaster Recovery](#). *npj Urban Sustainability*, Vol. 4, Issue 1, pp. 2. DOI: 10.1038/s42949-023-00139-4.
- [J.7] **Park, S.** and Kim, J. (2023). [Environmental Implications of Autonomous Demand Responsive Bus in Microscopic Traffic Simulation](#). *Chemical Engineering Transactions*, DOI: 10.3303/CET23106100.

- [J.6] **Park, S.** and Kim, J. (2022). [Impact Assessment of an Autonomous Demand Responsive Bus in a Microscopic Traffic Simulation](#). *The Journal of the Korea Institute of Intelligent Transport Systems*, Vol. 21, Issue 6, pp. 70-86. DOI: 10.12815/kits.2022.21.6.70.
- [J.5] Kim, J., **Park, S.**, Kim, K., Lee, S. (2019). [Environmental Implications on Energy and Emissions of the use of a Connected and Autonomous Vehicle Environment](#). *Chemical Engineering Transactions*, Vol. 72, pp. 97-102. DOI: 10.3303/CET1972017.
- [J.4] Lee, S., Heydecker, B. G., Kim, J., and **Park, S.** (2018). [Stability analysis on a dynamical model of route choice in a connected vehicle environment](#). *Transportation Research Part C: Emerging Technologies*, Vol. 94, pp. 67-82. DOI: 10.1016/j.trc.2017.10.019.
- [J.3] **Park, S.**, Kim, J., Lee, S., and Hwang, K. (2017). [Experimental Analysis on control constraints for connected vehicles using Vissim](#). *Transportation Research Procedia*, Vol. 21, pp. 269-280. DOI: 10.1016/j.trpro.2017.03.097.
- [J.2] **Park, S.**, Kim, J., Na, S., and Lee, S.. (2016). [Empirical analysis on radio communication range and vehicle ratio in V2X environment](#). *The Journal of the Korea Institute of Intelligent Transport Systems*, Vol. 15, Issue 5, pp. 29-41. DOI: 10.12815/kits.2016.15.5.029.
- [J.1] Kim, J., **Park, S.**, Na, S., and Lee, S. (2015). [Sustainable Highway Operations: High Occupancy Lane and Movable Lane](#). *International Journal of Transportation*, Vol. 3, Issue 3, pp. 1-10. DOI: 10.14257/ijt.2015.3.3.01.
- [T.2] **Park, S.** (2024). [CRITICAL TRANSITIONS OF POST-DISASTER RECOVERY VIA DATA-DRIVEN MULTI-AGENT SYSTEMS](#) (Doctoral dissertation, Purdue University Graduate School).
- [T.1] **Park, S.** (2017). [Traffic flow analysis under a connected and autonomous environment through market penetration rates](#) (Master Thesis, University of Seoul, Seoul, Republic of Korea).

PRESENTATIONS AND CONFERENCES

C=CONFERENCE, PR=PRESENTATION

- [C.16] **Park, S.**, Sun, Y., Jiang, S., and Zhao, X. Developing a Dynamic Social Vulnerability Index for Public Health (DSVI-PH) Using Location-based Data in Hurricane Beryl. In the 105th Transportation Research Board Annual Meeting, Washington D.C., 11-15, January 2026.
- [C.15] **Park, S.**, and Ukkusuri, S. V. (2025). Identifying transferable county-level regional interdependence of the post-disaster recovery process using causal discovery. At the US-Korea Conference 2025. Atlanta, GA, 8, August, 2025.
- [C.14] **Park, S.**, Sun, Y., Jiang, S., and Zhao, X. (2025). Developing a Dynamic Social Vulnerability Index for Public Health (DSVI-PH) to Support Emergency Response. At the 50th Annual Natural Hazards Workshop. Broomfield, CO, 16, July, 2025.
- [C.13] **Park, S.** (2025). Integrating Data to Map Human Mobility in Disasters. At the 50th Annual Natural Hazards Workshop. Broomfield, CO, 14, July, 2025.
- [C.12] Lee, S., **Park, S.**, Benedict, B. The structure of institutional and emergent social support networks in long-term disaster recovery: A case of Hurricane Harvey. At INSNA Sunbelt 2025, Paris, France, 2, June, 2025.
- [Pr.11] **Park, S.** (2025). Critical Transitions of Post-Disaster Recovery Via Data-Driven Multi-Agent Systems. At Gyeonggi Research Institute. Ui-wang, South Korea, 15, May 2025.
- [Pr.10] **Park, S.** (2025). Identifying transferable county-level regional interdependence of the post-disaster recovery process using causal discovery. At AI and Cities: An International Forum for Innovation and Collaboration. Gainesville, FL, 9-10, May 2025.
- [C.11] **Park, S.**, **Sun, Y.**, **Jiang, S.**, and **Zhao, X.** (2025). Developing a Dynamic Social Vulnerability Index for Public Health (DSVI-PH) to Support Emergency Response. At the National Hurricane Conference 2025. New Orleans, US, 16, April 2025.
- [C.10] **Park, S.**, **Sun, Y.**, **Jiang, S.**, and **Zhao, X.** (2025). Developing a Dynamic Social Vulnerability Index (DSVI) for Public Health Using Location-based Data In Hurricane Beryl. At South Eastern Regional Conference. Orlnado, US, 15, March 2025.
- [Pr.9] **Park, S.** (2025). Developing a Dynamic Social Vulnerability Index (DSVI) for Public Health Using Location-based Data In Hurricane Beryl. At Data for Good at Meta. Online, US, 11, March 2025.
- [Pr.8] **Park, S.** (2025). Tips for Academic Writing and Job Applications. At Incheon National University. Incheon, South Korea, 7, January 2025.
- [Pr.7] **Park, S.** (2025). Critical Transitions in Modeling Post-disaster Recovery Process. At the University of Seoul. Seoul, South Korea, 6, January 2025.
- [Pr.6] **Park, S.** (2025). Critical Transitions in Modeling Post-disaster Recovery Process. At the Seoul Institute. Seoul, South Korea, 6, January 2025.
- [Pr.5] **Park, S.** (2024). Post-disaster Recovery Process in the Era of Human-induced Climate Change. At the University of Seoul. Seoul, South Korea, 18, December 2024.

[Pr.4] **Park, S.** (2024). Post-disaster Recovery Process in the Era of Human-induced Climate Change. At Florida State University. Tallahassee, FL, 6, December 2024.

[C.9] **Park, S.,** Xue, J., and Ukkusuri, S. V. (2024). Finding Critical Transitions of the Post-disaster Recovery Using the Sensitivity Analysis of Agent-based Models. In the 103rd Transportation Research Board Annual Meeting, Washington D.C., 8-12, January 2024.

[C.8] **Park, S.** and Kim, J. (2023). Environmental Implications of Autonomous Demand Responsive Bus in Microscopic Traffic Simulation. In International Conference on Low Carbon Asia and Beyond 2023. Online, 20, October 2023.

[Pr.3] Ukkusuri, S. V., **Park, S.,** and Mittal, S. (2023). Accelerating AI Based Data Analytics for the Recovery of Cities after Disasters (and Pandemics). In One Concern Disaster Resilience Seminar Series, Online, 20, March 2023.

[Pr.2] **Park, S.** and Ukkusuri, S. V. (2023). Recovery of Human Mobility Patterns and Economic Impacts Due to Climate Change Events. In Purdue Road School Poster Session, West Lafayette, 14, March 2023.

[C.7] Xue, J., **Park, S.** (Presenter), Mondal, W. U., Reia, S. M., and Ukkusuri, S. V. (2023). Supporting Post-disaster Recovery with Agent-based Modeling on Multilayer Socio-physical Networks. In the 102nd Transportation Research Board Annual Meeting, Washington D.C., 8-12, January 2023.

[C.6] **Park, S.,** Kim, J., Kim, K., and Lee, S. (2018). Environmental Analysis on Energy and Emissions in a Connected and Autonomous Vehicles Environment. In 4th International Conference on Low Carbon Asia and Beyond (ICLCA'18), Malaysia, 24-26, October 2018.

[C.5] **Park, S.,** Lee, S., and Kim, J. (2018). Avoidable Collision Controls under A Connected and Autonomous Vehicles Environment. In 18th International Conference Road Safety on Five Continents (RS5C 2018), Jeju Island, Republic of Korea, 16-18, May 2018.

[C.4] Lee, S., Heydecker, B. G., Kim, J., and **Park, S.** (2017). Stability Analysis on A Dynamical Model of Route Choice in A Connected Vehicle Environment. In *Proceedings of the 22nd International Symposium on Transportation and Traffic Theory (ISTTT)*, Chicago, Illinois, USA, 24-26, July 2017, pp. 720-737.

[C.3] **Park, S.,** Kim, J., Lee, S., and Hwang, K. (2016). Experimental Analysis on Control Constraints for Connected Vehicles Using Vissim. In *Proceedings of the International Symposia of Transport Simulation (ISTS) and the International Workshop on Traffic Data Collection and Its Standardisation (IWTDCS)*, Seoul, Republic of Korea, 6-8, July 2016.

[C.2] Lee, S., Kim, J., Na, S., and **Park, S.** (2016). Analysis of Speed Limit Control in V2X Environment. In *Proceedings of ITS World Congress 2016*, Australia, 11-14, October 2016.

[C.1] **Park, S.,** Lee, S., and Kim, J. (2015). Traffic Simulation in V2X Environments with Vissim. In *8th PTV Asia-Pacific User Group Meeting (UGM) 2015*, Bangkok, Thailand.

[Pr.1] **Park, S.** and Lee, J. (2015). Transit Scheduling of Subway No. 9 to Reduce Congestion. Senior Project in Department of Transportation Engineering, Seoul, Republic of Korea.

RESEARCH SKILLS

-
- **Programming Languages:** Python, Java, and MATLAB
 - **Statistical Programs:** R, NLOGIT, Pylogit, and SAS
 - **Database Systems:** SQL, NoSQL (Studio3T), and PySpark
 - **Data Science & Machine Learning:** Pytorch, Gym, PyMC, Pandas, Scikit-learn, Seaborn, Causal-discovery Toolbox, DoWhy, econml, H2O, and Auto-sklearn
 - **GIS tool:** QGIS, PostgreSQL, and Geopandas (Python)
 - **Agent-based simulation:** Mesa (Python), MATSim (Java), Netlogo (Java), and Repast Symphony (Python, Java)
 - **Transportation Simulation:** EMME/4, TransCAD, VISSIM7+, TSIS-CORSIM (T7F), HCS, HSS, and SUMO
 - **Optimization:** GAMS, CPLEX, TOMLAB, and Pyomo
 - **Writing:** Word and Latex
 - **OS:** Windows and Ubuntu

HONORS AND AWARDS

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- **Honorable Mention - Postdoctoral Award** May 2025
AI and Cities: An International Forum for Innovation and Collaboration
 - **The Best Volunteer Award** April 2025
Korean-American Scientists and Engineers Association - Gainesville Chapter
 - **Winter 2025 News Letter Spotlight** February 2025
Public Health Extreme Events Research (PHEER)
 - **Post Doctoral Spotlight** August 2024
University of Florida Transportation Institute (UFTI)

• Erin Flanigan Award (\$385) <i>Purdue University</i>	2023
• Grand prize for Improving contribution of municipal government <i>Seoul Institute</i>	2017
• Fellowships for accelerated bachelor and master’s program (\$ 5,600) <i>University of Seoul</i>	2016, 2017
• The Certificate Awarded by Minister of Land and Infrastructure <i>University of Seoul</i>	2015
• Scholarship for Excellent Natural Sciences and Engineering (\$ 5,600) <i>University of Seoul</i>	2014, 2015
• Scholarship for Excellent Achievement (\$ 1,400) <i>University of Seoul</i>	2013
• Scholarship for Excellent Achievement (\$ 2,800) <i>University of Seoul</i>	2010

LEADERSHIP EXPERIENCE

• President <i>Korean-American Scientists and Engineers Association (KSEA) - Gainesville Chapter</i>	July 2025 - August 2025 [🌐]
• Membership Director (MD) <i>Korean-American Scientists and Engineers Association (KSEA) - Gainesville Chapter</i>	Oct 2024 - June 2025 [🌐]
• YG Advisor <i>University of Florida Korean-American Scientists and Engineers Association - Young Generation (UFKSEA-YG)</i>	Oct 2024 - May 2025 [🌐]
• Vice President <i>Institute of Transportation Engineers - Student Chapter at Purdue</i>	Aug 2022 - Aug 2023 [🌐]
• Vice President <i>Purdue Korean Association - Civil Engineers</i>	Aug 2022 - Aug 2023
• President <i>Purdue Korean Association - Civil Engineers</i>	Aug 2021 - Aug 2022

SERVICES

• Local Chair in the 2025 National Math and Science Competition <i>Korean-American Scientists and Engineers Association - Gainesville Chapter</i>	Apr. 5 2025
• Local Organizing Committee in the 2025 South East Regional Conference <i>Korean-American Scientists and Engineers Association</i>	Mar. 15 2025
• Moderator and Volunteer in Korea-Florida High Technology Forum <i>Neo City, FL.</i>	Jan. 13 2025 - Jan. 14 2025
• Volunteer in Location-Based Data Operations Group <i>Public Health Extreme Events Research (PHEER) network.</i> ◦ Working with Public Health Professor Prof. David Eisenman (UCLA, U.S.) and Mónica Rodrigues (University of Coimbra, Portugal).	Oct. 2024 - Apr. 2025
• Reviewer <i>Journals and Conferences</i> ◦ AMR20 and AMR50 in Transportation Research Board Annual Meeting (Jan. 2023 - Jun. 2025) ◦ Scientific Reports (SCIE) (Jun. 2025 - Current) ◦ Humanities and Social Sciences Communications (SSCI) (Apr. 2025 - Current) ◦ International Journal of Urban Science (SSCI) (Feb. 2025 - Current) ◦ Natural Hazards Review (SSCI) (Feb. 2025 - Current) ◦ Sustainable Communities (Dec. 2024 - Current)	
• Research Team Volunteer <i>AMR20 - Disaster Response, Emergency Evacuations, and Business Continuity in Transportation Research Board Annual Meeting</i>	Jan. 2023 - Jun. 2025

ADDITIONAL INFORMATION

Language: Korean (Fluent) and English (Fluent)

REFERENCES

1. **Dr. Satish V. Ukkusuri**
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2. **Dr. Amy Kim**
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Relationship: Postdoc Supervisor
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Relationship: Previous Postdoc Supervisor
4. **Dr. Seungyoon Lee**
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Relationship: Thesis Committee Member
5. **Dr. Seungjae Lee**
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Relationship: Master's Degree Advisor